PS-31 Waste Segregation at Source: CNN Classification of Dry Waste

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PS No. and Title: PS-31, Waste Segregation at Source | **GitHub:** https://github.com/KS-KARTHIK-05/CBL

1. Problem and Objective

Smart bins and material-recovery facilities must separate dry waste into recyclable streams without manual handling; every mis-sorted item loses recyclable material or contaminates a bale. The task is to classify a photo of a single waste item as glass, paper, cardboard, plastic, metal or trash. Because *trash* is only 5.4 % of the data, success is judged by **test macro-F1** (all classes weigh equally) and **per-class recall and F1, especially for trash**, with accuracy reported alongside.

2. Dataset and Pre-processing

Source: TrashNet, *dataset-resized* (github.com/garythung/trashnet), full set of 2,527 RGB images (512×384): glass 501, paper 594, cardboard 403, plastic 482, metal 410, trash 137 (imbalance 4.3 : 1). **Audit:** MD5 hashing found 3 identical photos filed under two classes (e.g. metal91 = glass115); all 6 copies were removed, leaving **2,521 images**. 19 near-duplicate pairs (same object shot twice) were kept within one split. **Split:** stratified, group-aware 70/15/15, seed 42: **1,766 train / 377 validation / 378 test** (trash 96/21/20), saved in *results/split.csv*. **Pre-processing:** resize to 224×224, ImageNet mean/std normalisation. **Augmentation (training only):** rotation ±20°, random resized crop (60–100 % area), horizontal/vertical flips, mild colour jitter. **Imbalance:** inverse-frequency class weights in the loss (trash 3.07, paper 0.71), checkpoint selection on validation macro-F1, and an ablation without the weights.

3. Methodology

Three CNNs were compared. **(A) WasteNet**, designed and trained from scratch: five stages of two 3×3 conv–BatchNorm–ReLU layers + 2×2 max-pool (32 → 384 channels, 224 → 7 px), global average pooling, dropout 0.4 and a linear layer (3.39 M parameters). **(B) ResNet-50** (23.5 M) and **(C) EfficientNet-B0** (4.0 M), pre-trained on ImageNet: the 1000-way head was replaced by dropout 0.3 + linear(6), and **all layers were fine-tuned (none frozen)**, the new head with a 10× higher learning rate. Loss: class-weighted cross-entropy with label smoothing 0.1. Optimiser: AdamW, one-cycle schedule (15 % warm-up, cosine decay), mixed precision on an RTX 4050 GPU. The weights with the best *validation* macro-F1 were kept, with early stopping on that metric (Table 1).

Table 1. Hyper-parameters (batch size 32 and input 224×224 for all models).

Model	Initialisation	Peak LR (backbone / head)	Weight decay	Max epochs	Early-stop patience
A. WasteNet	random	3e-3	0.05	60	15
B. ResNet-50	ImageNet	2e-4 / 2e-3	0.01	25	10
C. EfficientNet-B0	ImageNet	5e-4 / 3e-3	0.01	25	10

4. Results

All numbers are from the held-out **test set** (378 images, 20 trash). The model selected on *validation* macro-F1, **EfficientNet-B0**, reaches **accuracy 0.95 and macro-F1 0.95**, with trash recall 0.95 (19 of 20); per-class F1: glass 0.96, paper 0.96, cardboard 0.96, plastic 0.93, metal 0.94, trash 0.93. ResNet-50 scored slightly higher on test, but selection must use validation data and the CIs overlap.

Table 2. Test-set results (95 % bootstrap CI in brackets; CPU latency for one image on 4 threads).

Model	Accuracy	Macro-F1 [95% CI]	Trash P	Trash R	Trash F1	CPU ms	Train min
A. WasteNet (scratch)	0.82	0.80 [0.75, 0.84]	0.49	0.90	0.63	38.9	5.8
B. ResNet-50	0.97	0.97 [0.94, 0.98]	0.95	0.95	0.95	45.8	3.6
C. EfficientNet-B0 (selected)	**0.95**	**0.95 [0.92, 0.97]**	**0.90**	**0.95**	**0.93**	**18.5**	**3.0**
C, unweighted loss (ablation)	0.96	0.95 [0.93, 0.98]	0.95	0.95	0.95	18.5	2.7

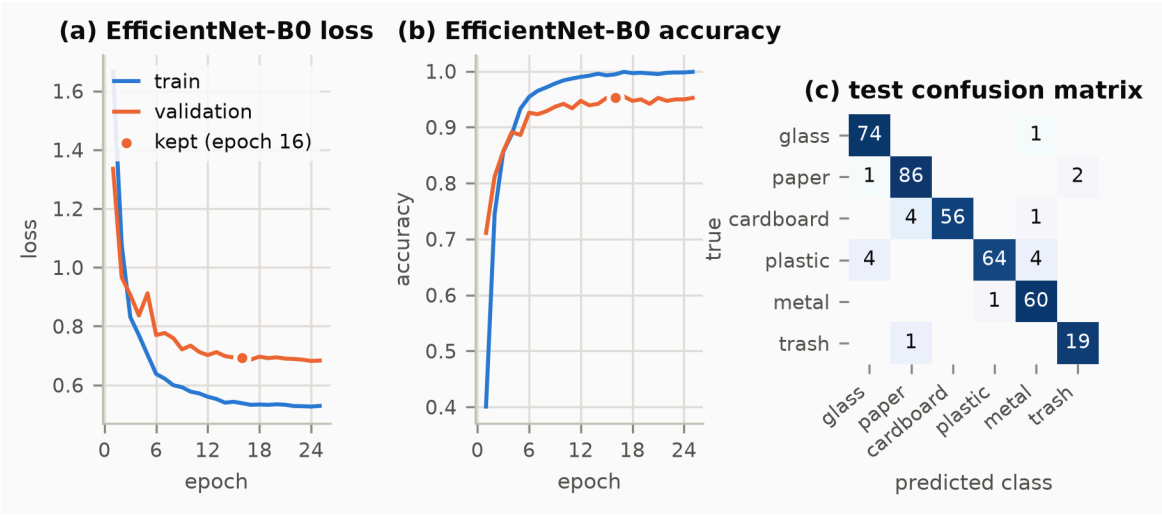


Fig. 1. EfficientNet-B0: (a) training and validation loss and (b) accuracy per epoch, with the dot marking the kept epoch (16); (c) test confusion matrix (counts).

5. Analysis and Discussion

Confusions (Fig. 1c): plastic is the hardest class (recall 0.89; 4 → glass, 4 → metal), then cardboard (0.92; 4 → paper). These are material look-alikes: PET bottles share shape and gloss with glass (Fig. 2c), lids resemble cans (Fig. 2d), white paperboard looks like paper. **Trash errors:** one of 20 was missed, a printed tea sachet predicted as paper (Fig. 2a). Trash is defined by recyclability rather than material, and in a t-SNE of the features this sachet lies inside the paper cluster. The 2 false trash alarms show a clear plastic bag that TrashNet labels *paper* (Fig. 2b), i.e. label noise. Grad-CAM shows correct trash predictions focus on the wrapper, not the background. **Fitting:** pre-trained models reach about 1.00 training vs. 0.95 validation accuracy with a flat validation loss, i.e. mild over-fitting contained by augmentation, dropout, weight decay and best-epoch selection; the scratch CNN under-fits (0.79 vs. 0.76). **Imbalance (specific challenge):** without class weights, EfficientNet-B0 gave trash F1 0.95 vs. 0.93 (1 vs. 2 false alarms), but validation macro-F1 0.94 vs. 0.95. These 1–3-image differences lie inside the CIs, so for pre-trained features (which form a separate trash cluster) the weights have no measurable effect; for the scratch CNN they raised trash recall (0.90) at the cost of precision (0.49).

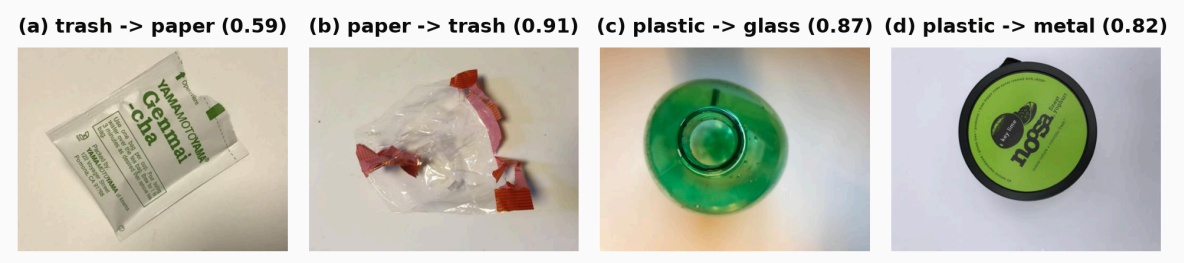


Fig. 2. Test errors of EfficientNet-B0 (true → predicted, confidence): (a) laminated tea sachet; (b) plastic bag labelled *paper* in TrashNet (label noise); (c) green PET bottle from above; (d) yoghurt-cup lid.

6. Real-world Relevance, Limitations and Ethics

The model could run in a camera-equipped smart bin or a conveyor pre-sorter: EfficientNet-B0 needs 4.0 M parameters (16 MB) and 18.5 ms per image on 4 CPU threads. It is not deployable yet: TrashNet shows one clean item on a white board, whereas real waste is dirty, cluttered and overlapping, and its US packaging may not represent Indian waste. Only 20 trash test images exist (trash-F1 CI 0.83–1.00). Since errors cost lost recyclables or contaminated bales, low-confidence items (wrong predictions average 0.59 confidence vs. 0.87 for correct ones) should go to manual sorting; bin cameras that may capture people should process images on the device without storing them.

7. Conclusion and Future Work

A fine-tuned EfficientNet-B0 sorts single waste items with **test macro-F1 0.95 and trash recall 0.95**, far above a CNN trained from scratch (macro-F1 0.80). Next, I would fine-tune it on photos taken inside a real bin (clutter, several items, dirt) and add a confidence-based reject option for uncertain items.

Dataset: G. Thung and M. Yang, "Classification of Trash for Recyclability Status," CS229 Project Report, Stanford University, 2016. **AI-assistance declaration:** Claude (Anthropic) assisted with the code and wording; all numbers come from the executed notebook.